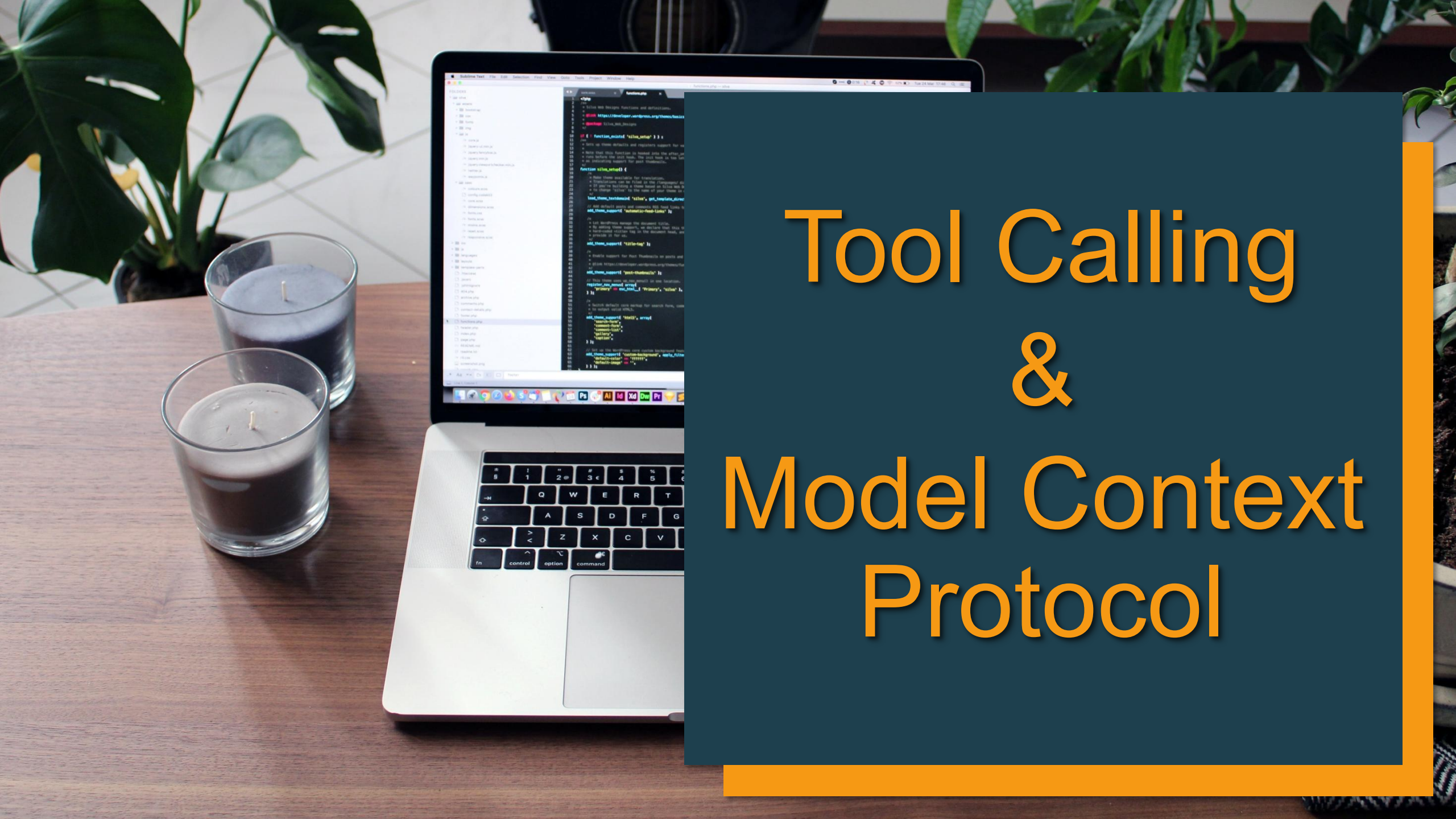

Tool Calling
&
Model Context
Protocol

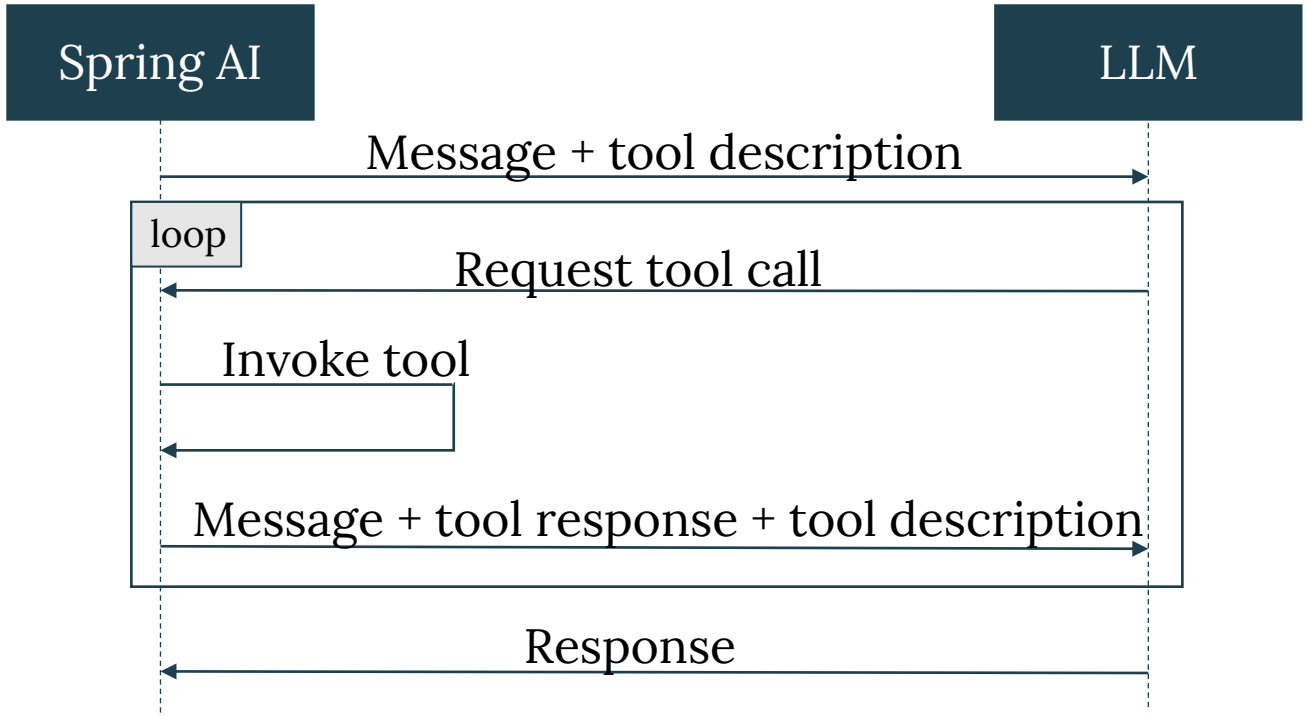
Tool Calling & MCP

- Provide access to custom APIs
  - Request information
  - Trigger operations


Tool Calling

Tool Calling / Function Calling

- Give model access to APIs



Code Samples

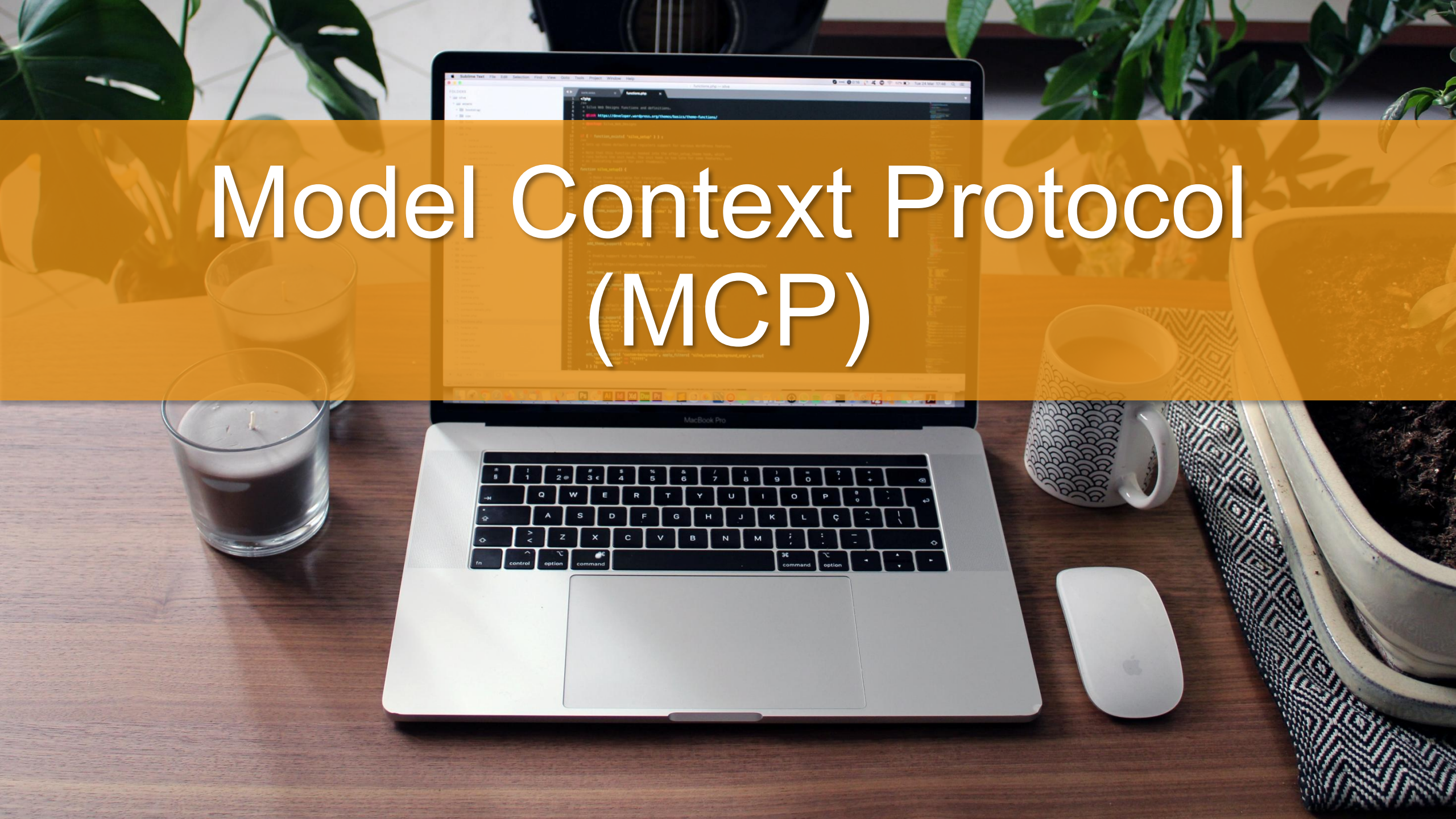
Manage Tools

- Problem
  - Tool definitions increase context size

Tool Search Tool

- Tell LLM to search for tools
  - Based on Advisor
- https://github.com/spring-ai-community/spring-ai-tool-search-tool

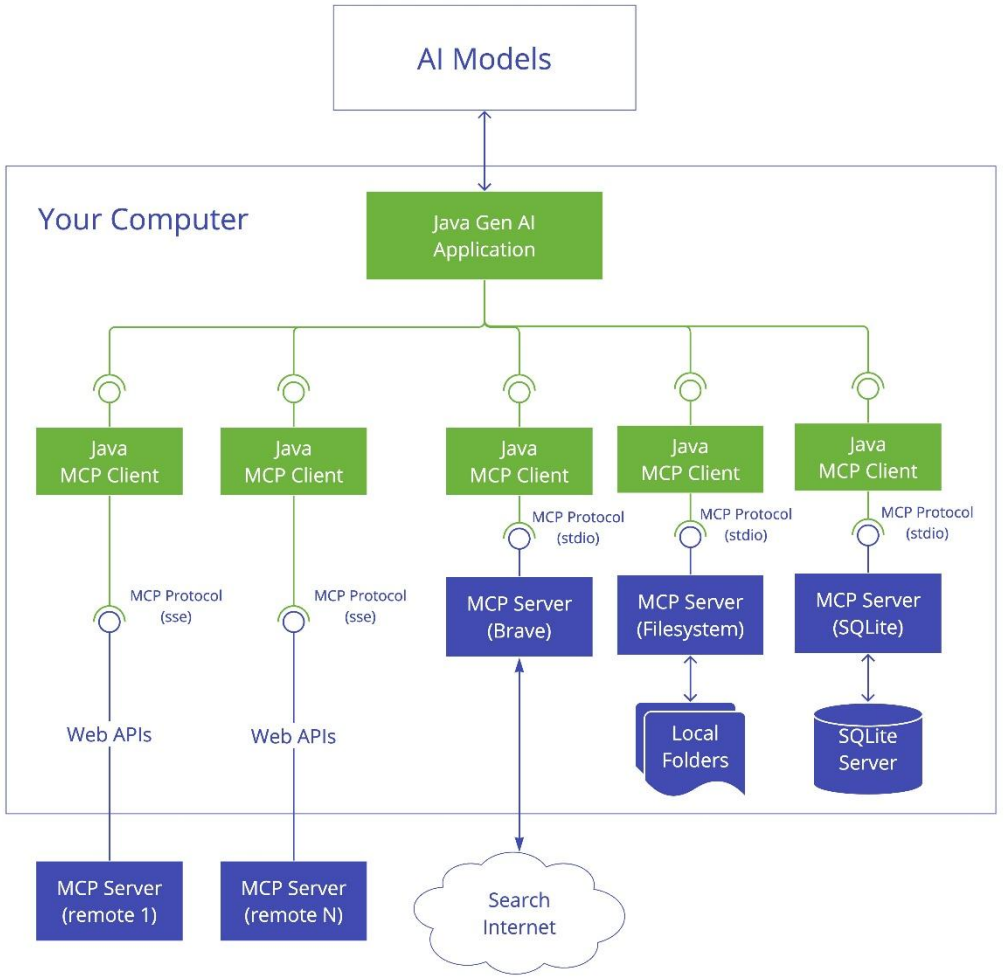
Code Samples


Model Context Protocol
(MCP)

MCP

- = tool calling for external services
- Standardized protocol
  - https://modelcontextprotocol.io/

MCP



Src: https://docs.spring.io/spring-ai/reference/api/mcp/mcp-overview.html

Dependencies

- Client
  - spring-ai-starter-mcp-client
  - spring-ai-starter-mcp-client-webflux
- Server
  - spring-ai-starter-mcp-server
  - spring-ai-starter-mcp-server-webmvc

MCP Annotations

- Server
  - @McpTool — tool call
  - @McpResource — access to a resource via URI
  - @McpPrompt — generates a prompt
  - @McpComplete — auto-completion

Code Samples

MCP Annotations

- Client
  - @McpLogging handles log messages
  - @McpSampling handles sampling request
  - @McpElicitation handles elicitation request
  - @McpProgress handles progress notifications

MCP Annotations

- Client
  - @McpToolListChange handles tool list change
  - @McpResourceListChange handles res. list change
  - @McpPromptListChange handles prompt list change


Security

MCP Security

- Spring AI Community project
- Based on Spring Security
  - Oauth 2.0
  - API key-based

MCP Security

- Documentation
  - https://docs.spring.io/spring-ai/reference/2.0/api/mcp/mcp-security.html
- Github
  - https://github.com/spring-ai-community/mcp-security


Questions